Chapter 42 — Multi-Account & Account Switching

> **Part V — Shared & Global State** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 22 (session/logout), Ch 20 (per-account token storage), Ch 19 (tokens), Ch 3 (SSOT/per-account data), Ch 38/39 (shared/global state), Ch 24 (encryption). Generalizes single-session management to **multiple accounts**.
>
> **Reference surface in StreamCart:** a user with a personal and a work account (or family profiles) switching between them, with completely isolated carts, orders, wishlists, and settings — and fast switching without re-login.

42.1 Problem Definition and Real-World Context

Many apps support **multiple accounts** on one device — personal + work email, multiple social profiles, family members, or a reseller managing several stores. Each account has its **own** data (cart, orders, settings, tokens) that must be **completely isolated**, and switching must be **fast** (no re-login) and **leak-free**. This is the hard generalization of session management (Ch 22); naive approaches leak data between users — a serious privacy failure:

- **Cross-account data leakage.** Account state (cart, cache, tokens) not scoped per account → switching shows the previous account's data, or a shared singleton mixes them. A top privacy incident.
- **Re-login on every switch.** Clearing everything on switch (like logout) forces re-authentication each time — terrible UX for frequent switchers.
- **Global state tied to one account.** A `@Singleton` holder (Ch 39) that assumes one account breaks: the cart/session/unread-count must become **account-aware**.
- **Token confusion.** One token store (Ch 20) holding a single token can't represent multiple logged-in accounts; requests use the wrong account's token.
- **Partial switch.** Switching updates some components but not others (a stale ViewModel, an in-flight request for the old account) → mixed data.

The architecture: a **current-account** concept driving **per-account data scoping** (every account-scoped store keyed by `accountId`), **per-account token storage** (Ch 20), a **fast switch** that changes the current account and re-points account-aware state (not a full logout/login), **data isolation** (one account can't see another's), and correct **in-flight request** handling. This chapter is about **scoping everything by account and switching safely and fast**.

Three practical use cases

1. **Personal + work (StreamCart).** Switch between a personal shopping account and a work procurement account — separate carts, orders, payment methods, settings.
2. **Family profiles (OTT).** Multiple viewer profiles (each with its own watchlist/ history/recommendations) switched from a profile picker.
3. **Multi-tenant (B2B).** A reseller managing several stores, switching context while keeping each store's data isolated.

Why this is architectural

Multi-account defines **how all account state is scoped and isolated**, the **current- account** concept that drives data access, **fast vs full switch** semantics, and the **privacy boundary** between accounts — properties that determine whether user data stays isolated and switching is usable.

42.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Support multiple logged-in accounts on one device.
F2	Scope all account data (DB/caches/prefs/tokens) per account; complete isolation.
F3	Switch the current account fast (no re-login) and leak-free.
F4	Store per-account tokens securely (Ch 20); use the current account's token.
F5	Make global/shared state account-aware (cart, session, unread — Ch 38/39).
F6	Add/remove accounts; sign out one without affecting others.
F7	Handle in-flight requests/work scoped to an account across a switch.

Non-functional requirements

Attribute	Policy
Privacy	Zero cross-account data leakage; complete isolation.
UX	Fast switch; no re-login; clear current-account indication.
Security	Per-account secure token storage; encrypt sensitive per-account data (Ch 24).
Consistency	Account-aware state re-points atomically on switch.
Correctness	In-flight/background work tied to the right account.

Scope

- **In scope:** current-account concept, per-account data scoping (keying by accountId), per-account token storage, fast account switch, data isolation, account-aware global state, add/remove/sign-out-one, in-flight request scoping.
- **Out of scope:** single-session logout (Ch 22), token refresh (Ch 19), secure storage internals (Ch 20), the shared-cart mechanics (Ch 38 — reused), encryption (Ch 24).

Assumptions & constraints

- Per-account token storage (Ch 20); Room with `accountId` columns (Ch 3); global holders made account-aware (Ch 39). Fast switch = re-point, not re-login. `minSdk 24`.

42.3 System Design Fundamentals

- **Current account as the scoping key.** A single `AccountProvider` holds the **current** `accountId` (a `StateFlow`). *Every account-scoped data access is keyed by it.* Switching accounts = changing the current id; account-aware state re-reads for the new id. This is the core mechanism: **one current-account signal drives all per-account scoping.**
- **Per-account data scoping.** Every account-scoped store includes `accountId`:
  - **Room:** an `accountId` column on account-scoped tables; queries filter by the current account (`WHERE accountId = :current`). Observers use `flatMapLatest` on the current account so they auto-switch.
  - **DataStore/prefs:** per-account keys or per-account files (Ch 4).
  - **Tokens:** per-account secure entries (Ch 20) — a map `accountId → tokens`.
  - **Caches:** keyed by account (Ch 12); cleared per account.
- **Fast switch vs full logout.** A **switch** changes the current account and re-points account-aware state (instant, no network) — it does **not** clear the old account's data (it stays for when you switch back). **Logout of one account** clears *that* account's data + revokes *its* session server-side (Ch 22), leaving others intact.
- **Account-aware global/shared state (Ch 38/39).** The cart, session, unread count, and profile must be **functions of the current account**: `observeCart()` becomes `currentAccount.flatMapLatest { dao.observe(it) }`. A `@Singleton` holder stays a singleton but its *state* is account-scoped internally.
- **Data isolation (the privacy invariant).** Account A's queries must **never** return account B's rows — enforced by always filtering on `accountId` (and ideally a DB constraint/row-level discipline). Test this explicitly.
- **In-flight work scoping.** A request/sync started for account A must complete for A even if the user switches to B — tag work with its `accountId`; on apply, write to A's data, not "the current account." Background workers (Ch 30) carry the `accountId` in `inputData`.
- **Security (Ch 20/24).** Each account's tokens are stored securely and separately; sensitive per-account data is encrypted; signing out one account wipes its data + tokens and revokes server-side (Ch 22).

Core insight: **make a single current-account id the scoping key for all account state, key every store by `accountId`, store tokens per account, and switch by re-pointing account-aware state (fast, isolated) — never leaking one account's data into another's view.**

42.4 High-Level Architecture



Components & ownership

- `AccountProvider` holds the current account id; the single scoping signal.
- **Per-account stores** (Room/tokens/prefs/caches) keyed by `accountId`.
- **Account-aware state** (cart/session) re-points via `flatMapLatest(current)`.
- **AuthInterceptor** uses the current account's token (Ch 19/20).
- **Background work** carries its `accountId`.

Direction: switch → change current account → account-aware state re-reads for the new id; all stores filter by account.

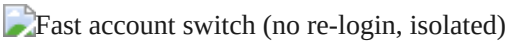
42.5 Detailed Component Design


42.5 Detailed Component Design

Component	Responsibility	Note
`AccountProvider`	current id + account list; switch/add/sign-out	the scoping key (Ch 39 global state)
per-account tokens	secure, separate per account	Ch 20 (map accountId → tokens)
account-scoped Room	`accountId` column; filter by current	Ch 3; `flatMapLatest` to auto-switch
account-aware state	cart/session functions of current account	Ch 38/22
sign-out-one	clear that account's data + revoke	Ch 22 (not all accounts)

42.6 End-to-End Data Flow

Fast account switch (no re-login, isolated)


Fast account switch (no re-login, isolated)

Sign out ONE account (others intact)


Sign out ONE account (others intact)

42.7 Complete Kotlin Implementation

Package: `com.streamcart.core.account`.

42.7.1 AccountProvider (current account + list)

```
@Singleton
class AccountProvider @Inject constructor(
    private val store: DataStore<Preferences>,          // persists current id + account list
(Ch 4)
    private val tokens: AccountTokenStore,              // per-account tokens (Ch 20)
    @ApplicationScope scope: CoroutineScope,
) {
    val currentId: StateFlow<String?> = store.data
        .map { it[stringPreferencesKey("current_account")] }
        .stateIn(scope, SharingStarted.Eagerly, null)

    val accounts: StateFlow<List<Account>> = store.data
        .map { parseAccounts(it) }.stateIn(scope, SharingStarted.Eagerly, emptyList())

    fun currentIdValue(): String = currentId.value ?: error("no account")

    suspend fun switchTo(accountId: String) {           // FAST: just change the current id
        require(tokens.has(accountId)) { "account not logged in" }
        store.edit { it[stringPreferencesKey("current_account")] = accountId }
        // account-aware state re-reads via flatMapLatest(currentId) — no clearing, no re-login
    }

    suspend fun add(account: Account, accountTokens: Tokens) {   // after a login
        tokens.put(account.id, accountTokens)           // per-account secure storage (Ch 20)
        store.edit { persistAccount(it, account); it[stringPreferencesKey("current_account")] =
account.id }
    }

    suspend fun signOut(accountId: String, revoke: suspend (String) -> Unit, clearData: suspend
(String) -> Unit) {
        revoke(accountId)                               // revoke THIS session server-side (Ch
22)
        clearData(accountId)                            // clear THIS account's data only
        tokens.remove(accountId)
        store.edit { removeAccount(it, accountId); switchToAnotherOrNone(it, accountId) }
    }
}
```

42.7.2 Per-account token store (Ch 20)

```
// Keyed by accountId; each account's tokens encrypted via Keystore (Ch 20).
class AccountTokenStore @Inject constructor(private val secure: SecureStore) {
    suspend fun put(accountId: String, t: Tokens) = secure.put("tokens:$accountId", json(t))
    suspend fun get(accountId: String): Tokens? = secure.get("tokens:$accountId")?.let(::parse)
    suspend fun has(accountId: String) = secure.get("tokens:$accountId") != null
    suspend fun remove(accountId: String) = secure.remove("tokens:$accountId")
}
// AuthInterceptor (Ch 19) reads the CURRENT account's token:
class TokenProviderImpl @Inject constructor(private val accounts: AccountProvider, private val
tokens: AccountTokenStore) : TokenProvider {
    override fun currentAccessToken(): String? = runBlocking { accounts.currentId.value?.let {
tokens.get(it)?.access } }
}
```

42.7.3 Account-scoped repository (auto-switches via flatMapLatest)

```
class CartRepositoryImpl @Inject constructor(
    private val dao: CartDao, private val accounts: AccountProvider, /* ... */
) : CartRepository {
    // Re-points to the current account automatically on switch (Ch 38).
    override fun observeCart(): Flow<Cart> =
        accounts.currentId.filterNotNull()
            .flatMapLatest { acct -> dao.observe(acct) }     // WHERE accountId = acct
(isolation)
            .map { it.toCart() }

    override suspend fun add(productId: String, qty: Int) {
        val acct = accounts.currentIdValue()                  // write to the CURRENT account
        dao.upsert(CartItemEntity(id = UUID.randomUUID().toString(), accountId = acct, productId
= productId, /*...*/))
    }
    suspend fun clearForAccount(accountId: String) = dao.clearAccount(accountId)   // sign-out-
one
}
```

42.7.4 Background work carries its accountId (Ch 30)

```
// A sync/upload worker must act on the account it was enqueued for, not "current".
val request = OneTimeWorkRequestBuilder<SyncWorker>()
    .setInputData(workDataOf("accountId" to accountId))       // bind to the account
    .build()
// doWork(): read inputData["accountId"] and operate on THAT account's data.
```

42.7.5 Module structure

```
:core:account  AccountProvider, Account, AccountTokenStore (Ch 20)
:data:*        account-scoped repos (accountId column, flatMapLatest current)
:core:network  TokenProviderImpl (current account's token, Ch 19)
```

42.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Switch during an in-flight request for A	result written to B	tag work/request with accountId; write to that account
Account-aware state reads mid-switch	brief wrong data	`flatMapLatest(currentId)` cancels old, reads new atomically
Background worker uses "current"	acts on wrong account	pass accountId in `inputData` (Ch 30)
Two accounts' tokens confused	wrong auth	per-account token store; current-account token
Concurrent switch + sign-out	inconsistent	serialize account ops; `AccountProvider` single writer
Query missing accountId filter	cross-account leak	always filter by accountId; test isolation

- **`flatMapLatest` on the current account id** makes account-aware observers switch **atomically**: on switch, the old account's collection is cancelled and the new one starts — no mixed data.
- **The isolation invariant is "every account-scoped query filters by accountId"** — a single missing filter is a cross-account leak; enforce by convention + tests (and ideally a DB constraint).

42.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Cross-account data shown	QA/privacy review	add accountId filter; `flatMapLatest`	isolated
In-flight result to wrong account	data in wrong cart	tag work with accountId	correct
Switch requires re-login	UX complaint	fast switch (re-point), per-account tokens	instant
One account's token expired	401	refresh that account (Ch 19); others unaffected	per-account
Sign out one wipes all	bug	clear only that account's data	others intact
Background worker on stale account	wrong data	accountId in inputData	correct
Account removed while selected	dangling current	switch to another/none	consistent
Device shared, no isolation	leak	encrypt per-account data (Ch 24); clear on sign-out	secure


42.9 Failure Scenarios and Recovery

> **Isolation is the privacy invariant:** the whole feature succeeds or fails on "account A never sees account B's data." Enforce `accountId` filtering everywhere and **test cross-account isolation explicitly** — it's the thing that breaks.

42.10 Scaling the Implementation

- **Level 1.** Single account (Ch 22) — most apps.
- **Level 2.** `AccountProvider` with current id; per-account tokens (Ch 20); account-scoped Room (accountId column).
- **Level 3 — production.** Fast switch (re-point via `flatMapLatest`); account-aware global/shared state (cart/session, Ch 38/22); sign-out-one; in-flight work tagging; encrypted per-account data (Ch 24).
- **Level 4 — large.** Many accounts; account switcher UX; per-account sync (Ch 29–33) and notifications routing (Ch 49); per-account feature flags/entitlements.
- **Level 5 — enterprise.** Managed/work profiles, MDM, account provisioning, cross-account audit, strict data-isolation guarantees (row-level security/ per-account encryption keys, Ch 24).

Avoid overengineering: single-account apps don't need this. Add multi-account only when users genuinely manage multiple accounts/profiles — it touches every store and adds real complexity.

42.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Scoping	accountId column + filter	separate DB per account	Column+filter simplest (one DB); separate DBs for strict isolation/encryption per account
Switch	Fast re-point (keep both)	full logout/login	Re-point for UX (frequent switchers); full switch if data must not persist
Tokens	Per-account store (map)	single token store	Per-account always for multi-account
State	Account-aware singleton (flatMapLatest)	re-create on switch	flatMapLatest (reactive, clean); re-create is heavier
Isolation	Filter by accountId	per-account encryption keys (Ch 24)	Filter for most; per-account keys for high-sensitivity
In-flight work	Tag with accountId	cancel on switch	Tag so A's work completes for A; cancel if ephemeral

Column+filter vs separate DB per account: keying one database by `accountId` (filter on current) is simplest and shares schema/migrations, but relies on discipline (every query filters) for isolation. A **separate database (and encryption key, Ch 24) per account** gives stronger isolation (a query physically can't cross accounts) at the cost of more files/complexity. Use column+filter by default; separate DBs/keys for high-sensitivity (fintech/enterprise).

Fast re-point vs full switch: fast switch keeps both accounts' data resident and just changes the current id (instant, no re-login) — great UX but both accounts' (possibly sensitive) data lives on-device simultaneously (encrypt it, Ch 24). A full logout/login-style switch avoids resident multi-account data but forces re-auth — rarely acceptable for a frequent switcher. Most apps choose fast re-point + encryption.

42.12 Security Considerations

- **Complete data isolation is a privacy requirement:** account A must never see B's data — filter every account-scoped query by `accountId`; consider per-account encryption keys (Ch 24) for strong isolation.
- **Per-account secure token storage** (Ch 20): each account's tokens encrypted and separate; the current account's token authorizes requests (Ch 19).
- **Sign out one account** revokes *that* session server-side (Ch 22) and wipes *that* account's data/tokens — leaving others intact; don't leak the signed-out account's residual data.
- **Shared device:** multi-account means multiple users' data on one device — encrypt sensitive per-account data (Ch 24); consider requiring auth (Ch 21) to switch to a sensitive account.
- **Backend authorizes per account:** the server must scope every response to the authenticated account (the client's current-account is a convenience; the backend enforces access).
- **In-flight work isolation:** a request for A must not write to B after a switch (tag by accountId) — else cross-account corruption.
- Client vs backend: the client scopes/presents per account; the backend is authoritative and must isolate data per authenticated account server-side.

42.13 Performance and Resource Optimization

- **Fast switch** = just change an id + `flatMapLatest` re-read — instant, no network; the big UX win.
- **Shared schema/one DB** (column+filter) avoids N databases' overhead; index `accountId` for fast filtering.
- **Account-aware singletons** (not per-account instances) keep memory bounded; `flatMapLatest` cancels the old account's collection.
- **Lazy per-account data:** load the current account's data; the inactive account's data stays on disk (not in memory).
- **Bound resident accounts** if many; evict least-recently-used account caches.
- **Profiling:** switch latency (should be ~instant), per-account DB size, memory with multiple accounts, isolation-query performance.

42.14 Testing Strategy

The critical test is **isolation**; plus fast switch and sign-out-one.

```
class MultiAccountTest {
    @Test fun `accounts are isolated - A never sees B data`() = runTest {
        accounts.add(Account("A"), tokensA); cart.add("p1", 1)        // A's cart
        accounts.switchTo("B"); accounts.add(Account("B"), tokensB)
        assertTrue(cart.observeCart().first().lines.isEmpty())         // B sees nothing of A
        accounts.switchTo("A")
        assertEquals(1, cart.observeCart().first().itemCount)          // A intact
    }

    @Test fun `switch is fast - no re-login, data re-points`() = runTest {
        accounts.add(Account("A"), tokensA); cart.add("p1", 1)
        accounts.add(Account("B"), tokensB); cart.add("p2", 1)
        accounts.switchTo("A"); assertEquals("p1",
cart.observeCart().first().lines[0].productId)
        accounts.switchTo("B"); assertEquals("p2",
cart.observeCart().first().lines[0].productId)
    }

    @Test fun `current account token used for requests`() = runTest {
        accounts.add(Account("A"), Tokens("tokA")); accounts.add(Account("B"), Tokens("tokB"))
        accounts.switchTo("A"); assertEquals("tokA", tokenProvider.currentAccessToken())
        accounts.switchTo("B"); assertEquals("tokB", tokenProvider.currentAccessToken())
    }

    @Test fun `sign out one clears only that account`() = runTest {
        accounts.add(Account("A"), tokensA); cart.add("p1", 1)
        accounts.add(Account("B"), tokensB); cart.add("p2", 1)
        accounts.signOut("B", revoke = {}, clearData = { cart.clearForAccount(it) })
        accounts.switchTo("A"); assertEquals(1, cart.observeCart().first().itemCount)   // A
intact
        assertFalse(tokenStore.has("B"))                                                 // B
gone
    }

    @Test fun `in-flight work writes to its account not current`() = runTest {
        // enqueue work for A; switch to B; worker writes to A
    }
}
```

What to test: cross-account isolation (the key one), fast switch, per-account token, sign-out-one, in-flight work tagging, encrypted per-account data (Ch 24). An instrumentation test for the account switcher UI.

42.15 Production Observability

- **Metrics:** account-switch frequency/latency (should be ~instant), number of accounts per device, cross-account-leak incidents (must be 0), sign-out-one success, per-account token-refresh rate, per-account data size.
- **Logs:** `account{op=switch/add/signout, count}` (ids hashed, no PII, Ch 28); never log which content belongs to which account.
- **Debugging:** previous account's data shows → missing `accountId` filter / not `flatMapLatest`; switch requires re-login → not using per-account tokens; in-flight result in wrong cart → work not tagged with accountId; sign-out wiped all → cleared globally not per account.
- **Alerting:** any cross-account-leak signal → privacy incident (highest severity).
- **Privacy:** multi-account = multiple users' data; redact; encrypt (Ch 24); honor per-account deletion.

42.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *What's the core of multi-account?* A current-account id that scopes all account data; switching changes it.
2. *How is data scoped?* Every account-scoped store keyed by `accountId`; queries filter by the current account.
3. *Fast switch vs logout?* Switch re-points to another (resident) account instantly; logout clears + re-auth.
4. *Per-account tokens?* A secure store keyed by accountId (Ch 20); current account's token authorizes.
5. *Isolation invariant?* Account A never sees B's data — every query filters by accountId.

Scoping/switch 6. *How do account-aware observers switch?* `flatMapLatest(currentId)` → cancel old, read new atomically. 7. *Make the cart account-aware?* `currentAccount.flatMapLatest { dao.observe(it) }` (Ch 38). 8. *Global singleton + multi-account?* Stays a singleton; its state is account-scoped internally. 9. *Column+filter vs DB-per-account?* Simple shared DB vs strong isolation/per-account encryption.

Correctness 10. *In-flight request during a switch?* Tag work with accountId; write to that account, not "current". 11. *Background worker?* accountId in `inputData` (Ch 30). 12. *Missing accountId filter?* Cross-account leak — the main bug.

Lifecycle 13. *Sign out one account?* Revoke that session (Ch 22) + clear that account's data/ tokens; others intact. 14. *Add an account?* Login → store per-account tokens → set current. 15. *Remove the current account?* Switch to another/none. 16. *Token expired for one account?* Refresh that account (Ch 19); others unaffected.

Security/privacy 17. *Shared device?* Encrypt per-account data (Ch 24); auth to switch to sensitive account (Ch 21). 18. *Who enforces access?* Backend per authenticated account; client scopes for UX. 19. *Strong isolation?* Per-account DB + encryption key (Ch 24).

Performance 20. *Why is switch fast?* Change an id + re-read; no network/re-login. 21. *Memory with many accounts?* Account-aware singletons; lazy per-account data; evict LRU.

Testing 22. *Key test?* Cross-account isolation. 23. *How test fast switch?* Add two accounts; switch; assert each sees its own data. 24. *How test sign-out-one?* Others intact; that account's tokens/data gone.

Production debugging 25. *Previous account's data shows?* Missing filter / not flatMapLatest. 26. *Re-login on switch?* Not using per-account tokens. 27. *Result in wrong account?* In-flight work not tagged.

Design-change challenges

1. **Add personal + work accounts with isolated carts.** accountId scoping + fast switch.
2. **Switch must be instant.** Per-account tokens + re-point (flatMapLatest).
3. **Cross-account leak found.** Enforce accountId filter everywhere; test isolation.
4. **Sign out one account.** Revoke + clear that account only (Ch 22).
5. **High-sensitivity isolation (fintech).** Per-account DB + encryption key (Ch 24).

Mock walkthrough (abridged)

Clarify (how many accounts, data per account, switch frequency, sensitivity?) → current-account id as scoping key → per-account stores (accountId column) + per-account tokens (Ch 20) → account-aware state via `flatMapLatest(current)` → fast switch (re-point, no re-login) → isolation invariant (filter everywhere) → in-flight work tagging → sign-out-one → encrypt per-account (Ch 24) → backend per-account authz → testing isolation.

42.17 Practical Exercises

1. **Beginner.** Add an `AccountProvider` with a current id + per-account token store (Ch 20); authorize requests with the current account's token. *Eval:* correct token per account.
2. **Intermediate.** Scope the cart (Ch 38) by accountId with `flatMapLatest(current)`; prove two accounts have isolated carts. *Eval:* A never sees B.
3. **Advanced.** Implement fast switch (no re-login) + sign-out-one (others intact) + in-flight work tagging. *Eval:* instant switch; isolated sign-out; correct account writes.
4. **Production debugging.** Fix a cross-account leak (missing accountId filter) and a switch that forces re-login. *Eval:* isolated; instant.
5. **System design challenge.** Add per-account encryption keys (Ch 24) for strong isolation and require biometric (Ch 21) to switch to a sensitive account. *Eval:* physically isolated; gated switch.

(Solutions withheld unless requested.)

42.18 Chapter Summary & Reference Sheet

Key decisions: drive all account state from a **single current-account id** (`AccountProvider`, a global holder, Ch 39); **scope every account store by** `accountId` (Room column + filter, per-account tokens Ch 20, per-account prefs/caches) for **complete isolation**; make **account-aware state re-point via** `flatMapLatest(current)` so **switching is fast** (change id, no re-login) and **leak-free**; **sign out one account** by revoking that session (Ch 22) + clearing *that* account's data/tokens only; **tag in-flight/background work with its accountId** so results go to the right account;

encrypt per-account sensitive data (Ch 24) on shared devices; the **backend authorizes per authenticated account**. The invariant: **account A never sees account B's data.**

Concepts: current-account scoping key, per-account data/token scoping, data isolation, fast switch vs logout, account-aware global/shared state, in-flight work tagging, column+filter vs DB-per-account, per-account encryption.

Common mistakes: missing `accountId` filter (cross-account leak); re-login on every switch (no per-account tokens); singleton assuming one account; in-flight work using "current" account (wrong writes); sign-out-one clearing all; unencrypted per-account data on shared devices; trusting the client's current-account for backend authz.

Checklist: ☐ current-account id (AccountProvider) ☐ accountId on every account store ☐ per-account secure tokens (Ch 20) ☐ account-aware state `flatMapLatest(current)` ☐ fast switch (re-point, no re-login) ☐ isolation: filter everywhere + test ☐ sign-out -one (others intact) ☐ tag in-flight/background work with accountId ☐ encrypt per-account sensitive data (Ch 24) ☐ backend authorizes per account.

Key APIs/patterns: `AccountProvider` (current id `StateFlow`), per-account `AccountTokenStore` (Ch 20), Room `accountId` column + `flatMapLatest(currentId)` (Ch 3/38), `TokenProvider` using current account (Ch 19), WorkManager `inputData` accountId (Ch 30), per-account encryption (Ch 24), sign-out-one (Ch 22).

References:

- (composes) Ch 22 Session, Ch 19/20 Tokens/Keystore, Ch 3 SSOT, Ch 38 Shared Cart, Ch 39 Global State, Ch 24 Encryption
- AccountManager (platform accounts): https://developer.android.com/reference/android/accounts/AccountManager
- Data isolation / privacy best practices: https://developer.android.com/privacy-and-security/security-tips

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; AccountProvider + per-account tokens + account-scoped repo + work tagging; concurrency (flatMapLatest switch, in-flight tagging); failure (leak/isolation/sign-out-one); 5 levels; column-vs-DB-per-account and fast-vs-full-switch trade-offs; security (isolation/per-account encryption/backend authz); perf (instant switch); isolation tests; observability (leak alert); 27 Q&A + 5 challenges. ✅
- **Expand later:** per-account notifications (Ch 49), per-account sync (Ch 29–33), managed/work profiles, per-account encryption keys (Ch 24).
- **Interview coverage:** exceeds ≥25 bar.